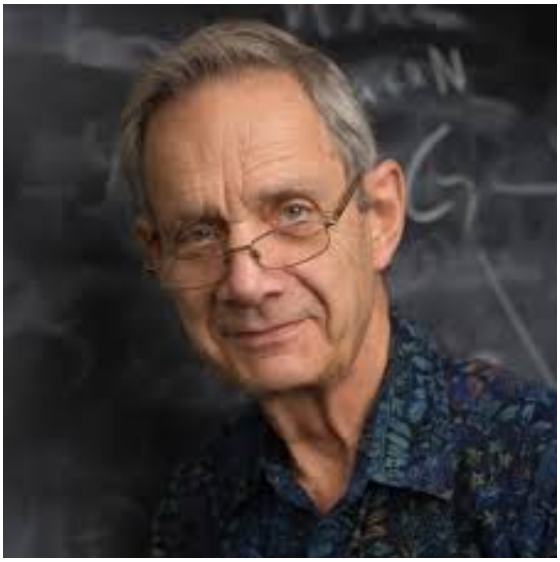
MATH621 - Algebraic Number Theory



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1 Overview

Class field theory(CFT) is the study of abelian extensions of global and local fields

Definition 1.1 A global field is a finite separable extension of $\mathbb{Q}$ or function field of a geometrically smooth curve over F_q. A local field is a finite extension of $\mathbb{Q}_p$ or function field of $F_q((t))$

we want to understand abelian extensions of K in terms of an invariant of K: the *idele class group*(some generalization of $\mathrm{Cl}(O_K)$)

$$C_K = \begin{cases} \mathbb{A}_k^\times / K^\times, K & \text{global} \\ K^\times, K & \text{local} \end{cases}$$

Why do we care?

Power reciprocity: The Legendre symbol $(n/p) = \begin{cases} 1 & n \text{ is a square } \mod p \\ -1 & \text{otherwise} \end{cases}$

Quadratic reciprocity: For p, q distinct odd primes $(p/q)(q/p) = 1$ if one of $p, q \equiv 1 \mod 4$, -1 if $p \equiv q \equiv 3 \mod 4$, or more succinctly $(p/q)(q/p) = (-1)^{\frac{(p-1)(q-1)}{4}}$. Also $(-1/p) = (-1)^{\frac{p-1}{2}}$, $(2/p) = (-1)^{\frac{p^2-1}{8}}$

Class field theory is a vast and conceptual generalization of this, it put quadratic reciprocity into context. CFT $\Rightarrow$ higher power reciprocity, e.g. cubic reciprocity: 2,7 are not cubic powers mod 61

A classical problem: $p = x^2 + ny^2$, when a prime p can be written as above

If $n = 1$, this holds iff $p \equiv 1 \mod 4$ or $p = 2$ iff p splits in $\mathbb{Q}(\sqrt{-1})$. CFT gives a complete solution to this for all n. See D. Cox "Primes of the form $x^2 + ny^2$"

If $n = 14$, then this holds iff $(-14/p) = 1$ and $(x^2 + 1)^2 - 8$ has root mod p

$K = \mathbb{Q}(\sqrt{-14})$, then this holds iff $p = \bar{P}P$ splits in K and P is principle iff(by CFT) $p = \bar{P}P$ splits in K and P splits in the Hilbert class field of K(H/K some specific finite abelian extension) iff p splits in H

"Reciprocity": whether a prime is principal is related to whether it splits in certain abelian extensions

"Class field": K is a number field, a modulus of K is a formal symbol $\mathfrak{m} = v_1^{e_1} \cdots v_k^{e_k}$. v_i's are places of K, and $e_i \in \mathbb{Z}_{\geq 0}$, satisfying: Complex places v don't appear in $\mathfrak{m}$, for real places v, $e_i = 0, 1$. e.g. $K = \mathbb{Q}$

Here a place is an equivalence class of absolute values, two absolute values are equivalent if they differ by a positive real power

The *ray class group* Cl_m is generated by fractional ideals coprime to $\mathfrak{m}$ modulo principal ideals generated by $f \in K^\times, f \equiv 1 \mod \mathfrak{m}$. In the number field case, Cl_m is finite which is no longer true for global fields with char>0

The usual class group $\mathrm{Cl}(\mathcal{O}_K)$ is where $\mathfrak{m} = 1$

Fact: Cl_m is always finite abelian

The narrow class group, corresponds to $\mathfrak{m}$ is the product of all real places. This is the quotient group of all fractional ideals modulo those principal ideals generated $x \in K^\times$ such that $v(x) > 0$ for all real places v, i.e., the totally positive $x \in K^\times$

CFT: there is a finite abelian ext K_m/K called *ray class field* of $\mathfrak{m}$

Example: $m = 1, K_m$ is the hilbert class field

K_m is uniquely characterized among finite abelian(Galois) extension over K such that whether prime p of K splits in K_m iff whether p has trivial image in Cl_m, for all p coprime to m

Theorem 1.2 (Generalized Kronecker-Weber theorem) Every finite abelian extension E/K is contained in K_m/K for sufficiently large $\mathfrak{m}$, one can choose $\mathfrak{m}$ such that its members are precisely the places of K that ramify in E

Example 1.3 If $v_1, \cdots, vk$ are the achmedian places of K that ramify in E, and $p_1 \cdots, p_k$ are unramified places, then $m = v_1, \cdots, v_k, p^{e_1}, \cdots, p_k^{e_k}$ for suff large $e_1, \cdots, e_k, E \subseteq K_m$

Artin isomorphism $\Psi : \mathrm{Cl}_m \to \mathrm{Gal}(K_m/K)$ has a concrete formula, $p \mapsto (p, K_m/K)$(well-definedness is nontrivial, called the Artin reciprocity). for every p coprime to m, know p is unramifed in K_m,

Recall: E/K is a finite Galois extension of global fields, suppose p is a prime of K that is unramifed in E, then $\forall P|p$, the arithmetic Frobenius element $\sigma = (P, E/K)$(Artin symbol) in $\mathrm{Gal}(E/K)$ is characterized by σ stablizes P, the action of $\sigma onk(P)$ as $x \mapsto x^q, q = |k(p)|$. $\{(P, E/K)|P\}$ runs through the primes of $P|p$ is a conjugacy class in $\mathrm{Gal}(E/K)$ called $(p, E/K)$. If $\mathrm{Gal}(E/K)$ is abelian, then $(p, E/K)$ is an element

Fact: For p of K unramified in E, $(p, E/K) = \{1\}$ iff p splits in E

The theory of ray CFT + Artin iso Ψ + K-W theorem gives the ideal theoretic formulation of global CFT

Adelic formulation in terms of $\mathbb{A}_k^\times / K^\times$ is cleaner. And it's easier to see functoriality in K that way

2 Class field theory over $\mathbb{Q}$

Application: Chebotarev density theorem

E/K is a finite Galois extension of number fields, $G = \mathrm{Gal}(E/K)$, forall p prime in K, unramified in E

Theorem 2.1 $\forall$ conjugacy classes C in G the set of p of K such that $(p, E/K) = C$ has density $|C|/|G|$ among all primes of K. In particular, there are infinitely many such primes p

applications in global Galois representation by density

consequence: p splits iff $(p, E/K) = \{1\}$, such primes constitute $1/|G|$ of all primes, thus infinitely many

Theorem 2.2 (Dirichlet theorem: primes in arithemetic progression) if $a, b \in \mathbb{Z}$, $(a, b) = 1$, exists infinitely many primees p in the arithemetic progression $a + b\mathbb{Z}$, e.g. $a = 1, b = 4$, infinitely primes $\equiv 1 \mod 4$

More application of CFT

Artin L funcitons: Let E/K be a finite Galois extension of number fields, $\rho : \mathrm{Gal}(E/K) \to GL_n(\mathbb{C})$, S finite primes including all ramified primes of K, define $L(\rho, s)$ for $\mathrm{re}(s) >> 0$, CFT $\Rightarrow$ meoromorphic extension to $\mathbb{C}$

Conjecture(Artin):..

Theorem 2.3 (Grumwold-Wang, local-global behavior of number fields, local-global principle for quadratic forms) A non-degenerate quadratic form over a number field K represents 0 over K(it=0 has sol over K) iff it represents 0 over K_v for all places v of K

CFT is the GL_1 case of the Langlands program

CFT for $\mathbb{Q}$(given by cyclotomic extension of $\mathbb{Q}$)

Review of cyclotomic extesions of $\mathbb{Q}$

K is a general field, $m \in \mathbb{Z}_{>0}$, the m-th cyclotomic extension of K is $K(\mu_m)$, μ_m is the roots of unity in $\bar{K}$, all roots would be simple, and is a cyclic group $\cong (\mathbb{Z}/m)^\times$ under multiplication, generator is primitive m-th root of 1, denoted ζ_m. $K(\mu_m) = K(\zeta_m)$, by definition also the splitting field of $x^m - 1$ over K, thus Galois. Obeservation: $\mathrm{Gal}(K(\zeta_m))$ embeds into $(\mathbb{Z}/m\mathbb{Z})^\times$, $\sigma \mapsto a, \sigma(\zeta_m)\zeta_m^a$, so the extension is abelian. Cyclotomic polynomials are $\Phi_m(x) = \prod_\zeta (x - \zeta) \in \mathbb{Z}[x]$, ζ runs primitive m-th roots of unity in $\mathbb{C}$

$\Phi_1 = x - 1$, $\Phi_2 = x + 1$, $\Phi_m = \frac{x^m - 1}{\prod_{d|m, d<m} \Phi_d(x)}$. $K(\zeta_m)/K$ is the spliting filed of Φ_m. $\deg(\Phi_m) = \phi(m) = |(\mathbb{Z}/m\mathbb{Z})^\times|$

$\alpha : \mathrm{Gal}(K(\zeta_m)/K) \to (\mathbb{Z}/m)^\times$ is an isomorphism iff Φ_m is irreducible

Theorem 2.4 (Gauss) Φ_m is irreducible in $\mathbb{Q}[x]$

Proof. Gauss's lemma, reduce to factorization mod p

□

Fact 2.5 (L washington sec2)

1. $\mathcal{O}_{\mathbb{Q}(\zeta_m)} = \mathbb{Z}[\zeta_m] \cong \mathbb{Z}[x]/\Phi_m(x)$
2. assume $m \equiv 2 \mod 4$(if $m \equiv 2 \mod 4$, then $\phi(m) = \phi(m/2) \Rightarrow \mathbb{Q}(\zeta_m) = \mathbb{Q}(\zeta_{m/2})$) prime p of $\mathbb{Q}$ is unramified in $\mathbb{Q}(\zeta_m)$ iff $p \nmid m$
3. formula for disc $\mathbb{Q}(\zeta_m)$

Lemma 2.6 $\forall p \nmid m$, $p \in (\mathbb{Z}/m)^\times$ is $(p, \mathbb{Q}(\zeta_m)/\mathbb{Q})$ the Frobenius element in $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$

Proof. Only need to prove σ fix P for $P|p$
Recall: Suppose E/K is finite separable extension of number fields, there is a way to explicitly factorize a prime p of K inside E(for almost all p), write $E = K(\alpha)$ such that $\alpha \in \mathcal{O}_E$, $\mathcal{O}_K[\alpha] \subseteq \mathcal{O}_E$ and $\mathcal{O}_k[\alpha] \otimes_{\mathcal{O}_E} E = E(\mathcal{O}_K[\alpha]$ is an order in $\mathcal{O}_E)$. Conductor: $f = \{x \in O_E | xO_E \subseteq O_K[\alpha]\}$, largest ideal of $\mathcal{O}_E$ that lies inside $\mathcal{O}_K[\alpha]$
Fact: for p prime of K, coprime to f, $p\mathcal{O}_E = \prod_{i=1}^{g} P_i^{e_i}$, $f(x)$ is the minimal polynomial of α in $\mathcal{O}_K[x]$, factorize over $k(p) = \mathcal{O}_K/p$, $\prod_{i=1}^{g} f_i^{e_i}$, f_i irreducible in $k(p)[x]$, P_i is a lift of $f_i(\alpha)\mathcal{O}_E + p\mathcal{O}_E$ $\mathcal{O}_E = \mathbb{Z}[\zeta_m]$, $\min(zeta_m/\mathbb{Q}) = \Phi_m$, $p\mathcal{O}_E = \prod P_i^{e_i}$, Φ_m in $F_p[x]$ factor as $\prod f_i^{e_i}$, P_i is a lift of $f_i(\zeta_m)$. Suppose p sends P_i to P_j, $i \neq j$, but p send P_i to lift of $f_i(\zeta_m^p) = h(\zeta_p)$, $h(\zeta_p) = (\tilde{f}(\zeta_m))^p$ in F_p implies $B_j \subseteq B_i$, contradiction!

□

Theorem 2.7 For $\mathbb{Q}$, a modulus is a symbol $\mathfrak{m} = \infty m$ or $\mathfrak{m} = m$ for some $m \in \mathbb{Z}_{>0}$, $Cl_{\mathfrak{m}}$ is the group of fractional ideals of $\mathbb{Q}$ coprime to m/principle ideals generated by $x \in \mathbb{Q}^\times$ such that x coprime to m, $x \equiv 1 \mod m$, $x > 0$ if $\mathfrak{m} = \infty \cdot m$

Exercise 2.8 When $\mathfrak{m} = \infty \cdot m$, then we have an iso $(\mathbb{Z}/m)^\times \to Cl_m$, $\forall p \nmid m$, the ray class group, $p \mapsto$ the class of the prime ideal (p) iso $(\mathbb{Z}/m)^\times/\{\pm 1\} \to Cl_m$, $\mathfrak{m} = m$, $\forall p \nmid m$, $p \mapsto$ the class of the prime ideal (p)

$E_m = \mathbb{Q}(\zeta_m)$, think of this as $Cl_{\infty m} \to \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$

This means that $\mathbb{Q}(\zeta_m)/\mathbb{Q}$ is the ray class field

Recall: this is also characterized by the following property: for almost all primes p, p splits in the ray class field iff p is trivial in the ray class group

for $p \nmid m$, p splits in m iff $(p, E_m/\mathbb{Q}) = 1$ iff p is trivial in $Cl_{\infty m}$

Note if $E \subseteq E_m$, then $E/\mathbb{Q}$ is again finte abelian extesion, and the composition $Cl_\infty m \to Gal(E_m/\mathbb{Q}) \overset{\text{restriction}}{\to} Gal(E/\mathbb{Q})$

Theorem[Kronecker-Weber, proof given by Hilbert, later simplified] Every finite abelian extension $E/\mathbb{Q}$ is over some $\mathbb{Q}(\zeta_m)$, can also choose m to be divisible only by those p that ramify in E

there is an elementary proof for the analogous result for $\mathbb{Q}_p$(local Kronecker-Weber, see Larry's book). Theorem true for all $\mathbb{Q}_p$ imply $\mathbb{Q}$(any non-trivial $E/\mathbb{Q}$ cannot be unramified everywhere)

CFT for $\mathbb{Q}$ is basically: Ψ and Kronecker-Weber theorem

Elegant proof for quadratic reciprocity: $p \neq q$ odd primes, and $q \equiv 1 \mod 4$, then $(p/q) = (q/p)$

there is a unique index 2 subgroup of $(\mathbb{Z}/q)^\times$, i.e. there is a unique quadratic extension K of $\mathbb{Q}$ inside E_q, we know that q is the only finite prime that ramifies in E_q, thus q is the only finite prime that ramifies in K, thus $K = \mathbb{Q}(\sqrt{q})(K \neq Q(\sqrt{-q})$ since $q \equiv 1 \mod 4, \mathrm{disc} = q$, not $4q$(2 also ramifies))

One can explicitly construct $\sqrt{q}$ inside $Q(\zeta_q)$(Gauss, see exercise in the notes). $(p/q) = 1$ iff p represents a square in $(\mathbb{Z}/q)^\times \cong \mathrm{Gal}(E_q/\mathbb{Q})$ iff p lies in the kernel of $\mathrm{Gal}(E_q/\mathbb{Q}) \to \mathrm{Gal}(K/\mathbb{Q})$ iff $(p, E_q/\mathbb{Q}) \to 1$ iff $(p, K/\mathbb{Q}) = 1$ iff p splits in K. $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{q}}{2}]$ contain $\mathbb{Z}[\sqrt{q}]$ with conductor $2\mathcal{O}_K$ iff $X^1 - q$ splits mod p iff $(q/p) = 1$

Exercise 2.9 Try other cases: prove $(p/q) = -(q/p)$ if $p \equiv q \equiv 3 \mod 4$

3 Ramification in $\mathbb{Q}(\zeta_m)$

$p \nmid m$ imply p unramified in $\mathbb{Q}(\zeta_m)$, $p\mathcal{O}_{\mathbb{Q}(\zeta_m)} = P_1^e \cdots P_g^e$, $f = [k(P_i) : k(p)]$, $efg = [\mathbb{Q}(\zeta_m), \mathbb{Q}] = \varphi(m)$. $e = 1$, f is the order of p in $(\mathbb{Z}/m\mathbb{Z})^\times$

Lemma 3.1 If $m = p^r$, $\mathbb{Q}(\zeta_m)/\mathbb{Q}$ is totally unramified at p, i.e. $p\mathcal{O}_{\mathbb{Q}(\zeta_m)} = P^{[\mathbb{Q}(\zeta_m):\mathbb{Q}]}$

Proof. Modulo p, $\Phi_m(X) = \frac{X^{p^r}-1}{X^{p^{r-1}-1}} = \frac{(X-1)^{p^r}}{(X-1)^{p^{r-1}}} = (X-1)^{\varphi(m)}$, thus $p\mathcal{O}_{\mathbb{Q}(\zeta_m)} = P^{\varphi(m)}$

□

In general, $m = np^r$, $p \nmid n$, $\mathbb{Q}(\zeta_m)$ is the composite $\mathbb{Q}(\zeta_n)\mathbb{Q}(\zeta_{p^r})$, and $\mathbb{Q}(\zeta_n), \mathbb{Q}(\zeta_{p^r})$ are linearly disjoint over $\mathbb{Q}$ since $\varphi(m) = \varphi(n)\varphi(p^r)$

$p \nmid m$, $\mathbb{Q}(\zeta_m)_P/\mathbb{Q}_p \cong \mathbb{Q}_p(\zeta_m)/\mathbb{Q}_p$, here $\mathbb{Q}(\zeta_m)_P = \mathbb{Q}_p(\zeta_m)$ is the composite $\mathbb{Q}_p\mathbb{Q}(\zeta_m)$

$D_P(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \cong \mathrm{Gal}(\mathbb{Q}_p(\zeta_m)/\mathbb{Q}_p)$, thus $\mathbb{Q}_p(\zeta_m)/\mathbb{Q}_p$ is unramified, and its degree is the order of p in $\mathbb{Z}/m\mathbb{Z}^\times$. Similarly, $m = p^r$, $\mathbb{Q}_p(\zeta_m)/\mathbb{Q}_p$ is totally unramified, and its degree is $e = \varphi(p^r)$

Fact 3.2 K local field, $E/K, F/K$ two finite Galois extensions. If E/K is unramified of degree f and F/K is totally ramified of degree e, then E, F are linear disjoint over K, $[E : K] = ef$, e is ramification index, f is residue extension degree

$\mathbb{Q}_p(\zeta_m)$ is the composite $\mathbb{Q}_p(\zeta_n)\mathbb{Q}_p(\zeta_{p^r})$, $e = \varphi(p^r)$, f is the order of p in $\mathbb{Z}/m\mathbb{Z}^\times$. $[\mathbb{Q}_p(\zeta_m) : \mathbb{Q}_p] = ef$. The Galois group is the direct product

$$\begin{array}{ccc} \mathrm{Gal}(\mathbb{Q}_p(\zeta_m)/\mathbb{Q}_p) & \longrightarrow & \mathrm{Gal}(\mathbb{Q}_p(\zeta_n)/\mathbb{Q}_p) \times \mathrm{Gal}(\mathbb{Q}_p(\zeta_{p^r})/\mathbb{Q}_p) \\ \uparrow \alpha_m & & \uparrow \\ (\mathbb{Z}/m)^\times & \longrightarrow & (\mathbb{Z}/n)^\times \times (\mathbb{Z}/p^r)^\times \end{array}$$

$\alpha_n : \mathrm{Gal}(\mathbb{Q}_p(\zeta_n), \mathbb{Q}_p) \to (\mathbb{Z}/n)^\times$ sends Frobenius element to p, and the subgroup generated by Frobenius element to the subgroup generated by p

α_{p^r} is an isomorphism because $[\mathbb{Q}_p(\zeta_{p^r}) : \mathbb{Q}_p] = \varphi(p^r) = |(\mathbb{Z}/p^r)^\times|$

$\mathbb{Q}_p^\times = p^{\mathbb{Z}} \times \mathbb{Z}_p^\times$, $\mathbb{Z}_p^\times = \lim_{\leftarrow n} (\mathbb{Z}/p^n)^\times$, $\mathbb{Z}_p^\times/(1 + p^r\mathbb{Z}_p) \cong (\mathbb{Z}/p^r)^\times$

Define a map $j_m : \mathbb{Q}_p^\times \to (\mathbb{Z}/m)^\times$

$$\begin{array}{ccccccc} \mathbb{Q}_p^\times & \cong & p^{\mathbb{Z}} & \times & \mathbb{Z}_p^\times \\ \downarrow j_m & & \downarrow p \mapsto px \mapsto x^{-1} \mapsto \text{natural projection} & & \downarrow \\ (\mathbb{Z}/m)^\times & \cong & (\mathbb{Z}/n)^\times & \times & (\mathbb{Z}/p^r)^\times \end{array}$$

$$\mathbb{Q}_p^\times \longrightarrow (\mathbb{Z}/m)^\times \longleftarrow \mathrm{Gal}(\mathbb{Q}_p(\zeta_m)/\mathbb{Q}_p)$$

$$\mathbb{Q}_p^\times \xrightarrow{\psi_m} \mathrm{Gal}(\mathbb{Q}_p(\zeta_m)/\mathbb{Q}_p)$$

ψ_m is continuous and surjective
Suppose $m' \in \mathbb{Z}_{\geq 1}$ divisible by m, then $\mathbb{Q}_p(\zeta_m) \subseteq \mathbb{Q}_p(\zeta_{m'})$

$$\begin{array}{ccccc} \mathbb{Q}_p^\times & \longrightarrow & (\mathbb{Z}/m)^\times & \longleftarrow & \mathrm{Gal}(\mathbb{Q}_p(\zeta_m)/\mathbb{Q}_p) \\ \| & & \downarrow & & \downarrow \\ \mathbb{Q}_p^\times & \longrightarrow & (\mathbb{Z}/m')^\times & \longleftarrow & \mathrm{Gal}(\mathbb{Q}_p(\zeta_{m'})/\mathbb{Q}_p) \end{array}$$

Take the inverse limit $\phi : \mathbb{Q}_p^\times \to \lim_{\leftarrow m} \mathrm{Gal}(\mathbb{Q}_p(\zeta_m)/\mathbb{Q}_p) = \mathrm{Gal}(\cup\, \mathbb{Q}_p(\zeta_m)/\mathbb{Q}_p) = \mathrm{Gal}(\mathbb{Q}_p^{cyc}/\mathbb{Q}_p)$.
The image is dense

Theorem 3.3 (local Kronecker-Weber theorem for $\mathbb{Q}_p$) Every finite abelian ext of $\mathbb{Q}_p$ is contained in some $\mathbb{Q}_p(\zeta_m)$. $\mathbb{Q}_p^{cyc} = \mathbb{Q}_p^{ab}$ is the maximal abelian extension(which is the union of all finite abelian extension) of $\mathbb{Q}_p$ in $\overline{\mathbb{Q}_p}$

One of the main goals of local CFT is: for every local field K, to construct a map $K^\times \to \mathrm{Gal}(K^{ab}/K)$(local Artin map) and study its behaviour

e.g. If L/K is a finite abelian extension, if restrict to L/K(finite abelian extension, ψ_L/K), ϕ_L/K is surj and $\ker \psi_L/K = Im(N_L/K : L^\times \to K^\times)$

Can characterize which subgroups of $K^\times$ are of the form $\ker \psi_L/K$ for some L

Local-global relationship: We have local Artin map $\psi_p : \mathbb{Q}_p^\times \to \mathrm{Gal}(\mathbb{Q}_p^{ab}/\mathbb{Q}_p)$ and the global Artin map $\Psi_m : (\mathbb{Z}/m)^\times \to \mathrm{Gal}(\mathbb{Q}\zeta_m/\mathbb{Q})$

$$\begin{array}{ccc} (\mathbb{Z}/m)^\times & \xrightarrow{\Psi_m} & \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \\ \uparrow{\scriptstyle j_{m,p}} & & \uparrow \\ \mathbb{Q}_p^\times & \xrightarrow{\psi_{p,m}} & \mathrm{Gal}(\mathbb{Q}_p(\zeta_m)/\mathbb{Q}_p) = D_p(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \end{array}$$

Definition 3.4 (Chevalley(finite ideles)) $\mathbb{A}_f^\times = \left\{(x_p) \in \prod_p \mathbb{Q}_p^\times | x_p \in \mathbb{Z}_p^\times \text{ for almost all } p\right\}$, ideles is $\mathbb{A}^\times = \mathbb{R}^\times \times \mathbb{A}_f^\times$

$\mathbb{Q}^\times$ diagonally sits in $\mathbb{A}^\times$, i.e. $x \mapsto (x_\infty, x_2, x_3, x_5, \cdots)$. Idelic global Artin map $\Psi_m : \mathbb{A}_f^\times \to Gal(\mathbb{Q}(\zeta_m)/\mathbb{Q}), (x_p) \mapsto \psi_{\infty,p} \prod \psi_{p,m}(x_p)$ which is a finite product since $\mathbb{Z}_p^\times$ elements go to 0

Here $\psi_{\infty,p}(x)$ is identity if x is positive, and σ_∞ which is $-1 \in \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) = (\mathbb{Z}/m)^\times$ if x is negative. So

$$(x_\infty, x_f) \mapsto \begin{cases} \Psi(x_f) & x_\infty > 0 \\ \sigma_\infty \Psi(x_f) & x_\infty < 0 \end{cases}$$

Exercise 3.5 This is actually a map $A^\times/\mathbb{Q}^\times \to \mathrm{Gal}(\mathbb{Q}^{cyc}/\mathbb{Q}) = \mathrm{Gal}(\mathbb{Q}^{ab}/\mathbb{Q})$

Global CFT: For any global field K, Artin map $\Psi : A_K^\times/K^\times \to \mathrm{Gal}(K^{ab}/K)$

Future plan:

review of profinite groups: allows us to talk about infinite Galois theory

State the main theorems in local CFT after reviewing some basics about local fields

Lubin-Tate theory: analogue of the local cyclotomic extensions, gives an explicit construction of K^{ab} for a local field K

Group cohomology

Use Group cohomology to fully prove local CFT

Global CFT

4 Profinite groups

Example of direct system:

1. $(\mathbb{N}, \geq)$
2. $(\mathbb{N}, |)$
3. G is a (topological)group, $I = \{\text{finite index(open) normal subgrps of } G\}$, $N \geq N'$ if $N \subseteq N'$
4. Fix a field extension E/K, $I = \{L/K \text{ finite Galois}\}$, $L \geq L'$ if $L \supseteq L'$. If L, L' are both finite Galois, then so is their composite

Let C be a category(sets, groups, rings, top groups, top spaces, top rings). A profinite group is a topological group isomorphic to a inverse limit of finite groups. A profinite group is Hausdorff and compact

Theorem 4.1 (Tychonoff's theorem) A product of compact spaces is compact

Theorem 4.2 G is a topological group, G is profinite iff it is Hausdorff compact and $1 \in G$ has a neighborhood basis consisting of open subgroups of G

Proof. $\Leftarrow$: Suppose $G = \lim_{\leftarrow} G_i$, $\forall i \in I$, $N_i = \ker(G \to G_i)$ give such a basis
$\Rightarrow$:

□

Remark 4.3 In a topological group, every open subgroup is closed(being the complement of the union of its other cosets). In a compact group, a closed subgroup is open iff it's of finite index(conjugates cover G, then by compactness), and every open subgroup is closed hence compact

Example 4.4 $\mathbb{Z}_p$ with topology given by the p-adic value, $p^n\mathbb{Z}_p$ form a neighborhood basis of 0 in $\mathbb{Z}_p$ consisting of open subgroups. $\mathbb{Z}_p^\times = \lim_{\substack{\leftarrow \\ n}} (\mathbb{Z}/m\mathbb{Z})^\times$ has $1 + p^n\mathbb{Z}_p$ as an open neighborhood basis of 0

Remark 4.5 A topological group is called locally profinte it's locally compact and Hausdorff and 1 has a neighborhood basis consisting of open subgroups. Equivalently, G has an open subgroup which is profinite

Example 4.6 $Q_p \overset{\text{open}}{\supseteq} Z_p$ is locally profinite since its not compact

Fact: A topological group is (locally) profinite iff it is Hausdorff, (locally compact) and totally disconnected

G is a topological group, I is the set of open normal finite index subgroups(Note: quotient top on G/N_i is discrete), so $\hat{G} = \lim_{\leftarrow} G/N_i$ defines a profinite group, called the profinite completion, and a natural continuous map $G \to \hat{G}$, it is an isomorphism iff G is profinite. For every continuous $G \to H$, where H is profinite, the it factors through $\hat{G} \to H$

Exercise: $\hat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$

Infinite Galois theory: E/K Galois(separable and normal), I is the subset consists of L/K which are finite Galois

Fact(easy): As abstract groups, $\mathrm{Gal}(E/K) = \lim_{\leftarrow} \mathrm{Gal}(L/K)$, define Krull topology on $\mathrm{Gal}(E/K)$ by profinite topology given by the right hand side

Theorem 4.7 (Galois correspondence) Sub-extensions L/K corresponds to closed subgroups of $\mathrm{Gal}(E/K)$, $L \mapsto \mathrm{Gal}(E/L)$, $H \mapsto E^H$, $E^H = E^{\bar{H}}$, finite extensions are in bijective correspondence to open subgroups, Galois extensions are in bijective correspondence to normal subgroups.

If L/K is Galois, then $\mathrm{Gal}(E/L)$ is normal in $\mathrm{Gal}(E/K)$, and $\mathrm{Gal}(L/K) \cong \mathrm{Gal}(E/K)/\mathrm{Gal}(E/L)$ as topological groups

Example 4.8 $\mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) \cong \lim_{\underset{n}{\leftarrow}} \mathrm{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) \cong \lim_{\underset{n}{\leftarrow}} \mathbb{Z}/n\mathbb{Z},\ \mathbb{Z} \to \hat{\mathbb{Z}},\ 1 \mapsto 1$ is the Frobenius element

remark: $G =$ profinite, S is dense iff image of S in each G_i is G_i, since G_i is discrete

$(\mathbb{Z}/m\mathbb{Z})^\times \cong \mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}),$ thus $\hat{\mathbb{Z}}^\times \cong \mathrm{Gal}(\mathbb{Q}^{cyc}/\mathbb{Q}) = \mathrm{Gal}(\mathbb{Q}^{ab}/\mathbb{Q})$ But $\mathbb{Q}_p^\times \to \mathrm{Gal}(\mathbb{Q}_p^{ab}/\mathbb{Q}_p)$ only has dense image

Exercise 4.9 G is locally profinite, and $\phi : G \to G'$ is continuous, G' is profinite, ϕ has dense image, then $\hat{G} \cong G'$

5 Local fields

A discrete valued field (K, v) is a surjective(normalized and exclude trivial valuations) function $v : K \to \mathbb{Z} \cup \{+\infty\}$ satisfying

1. $v(x) = +\infty \Leftrightarrow x = 0$
2. $v(xy) = v(x) + v(y)$ is a group homomorphism $K^\times \to \mathbb{Z}$
3. $v(x + y) \geq \min(v(x), v(y))$

Subring $\mathcal{O}_K = \{x \in K | v(x) \geq 0\}$ of K with $\text{Frac}(\mathcal{O}_K) = K$ and that it is a valuation ring(in fact a DVR, i.e. a PID with unique non-zero prime ideal), with the unique non-zero prime ideal $m_K = \{x \in K | v(x) > 0\}$, generated by the uniformizer π such that $v(\pi) = 1$. $\mathcal{O}_K^\times = \{x \in K | v(x) = 0\}$. For any $x \in K$, there is a unique n such that $\pi^{-n}x \in \mathcal{O}_K^\times$, $n = v(x)$, i.e. v can be recovered from $\mathcal{O}_K$, in fact, all discrete valuations v on K corresponds to DVR's $\mathcal{O} \subseteq K$ whose fraction field is K. $k = \mathcal{O}_K/m$ is the residue field, then is a natural topology on K, pick $\alpha \in \mathbb{R}$, $0 < \alpha < 1$, define absolute value on K, $K^\times \to \mathbb{R}_{>0}$, $x \mapsto \alpha^{v(x)} = |x|$. Discrete valuations correspond to non-Archimedean absolute values whose image is a discrete subgroup of $\mathbb{R}^\times/\sim$, making K into a metric space, whose topology is independent of α. In fact, $\mathcal{O}_K$ is open, and m^n, $n \geq 1$ form an open neighborhood basis of 0

Theorem 5.1 (Ostrowski's theorem) Every non-trivial absolute value on $\mathbb{Q}$ is equivalent to either the real absolute value $||_\infty$ or p-adic absolute values $||_p$. A field that is complete with respect to an Archimedean absolute value is(topologically and algebraically) $\mathbb{R}$ or $\mathbb{C}$

Note. An absolute value is a norm with $|xy| = |x||y|$. Two absolute values $||, ||_*$ are equivalent if $||_* = ||^c$ for some $c > 0$. The trivial absolute value is $|0| = \infty$ and 0 otherwise

Example 5.2 $K = \mathbb{Q}, v = v_p, \mathcal{O}_K = \mathbb{Z}_{(p)}, m = p\mathbb{Z}_{(p)}$ is locally profinite. $K = \mathbb{Q}_p, v = v_p, \mathcal{O}_K = Z_p$ is profinite

Definition 5.3 (R, m) is a local ring, its completion $\hat{R} = \lim_{\leftarrow n} R/m^n$. $\hat{R}$ is also local with unique max ideal $\hat{m} = \ker(\hat{R} \to R/m)$, $R \to \hat{R}$ is a natural local ring homomorphism, induce $R/m^n \cong \hat{R}/\hat{m}^n$, thus $\hat{\hat{R}} \cong \hat{R}$

We call (K, v) complete if $\mathcal{O}_K$ is complete as a local ring. and this is true iff K is complete in the metric sense

Definition 5.4 A non-archimedean local field is a discrete valued field (K, v) which is complete and has finite residue field. Archimedean local fields are $\mathbb{Q}, \mathbb{C}$ by Theorem 5.1

We use local fields to mean just non-archimedean local fields. In this case, $\mathcal{O}_K/m^n$ are discrete by exact sequences, so $\mathcal{O}_K$ is profinite, thus compact, K is locally profinite. Conversely, if (K, v) is a discrete valued field such that K is locally profinite(suffices to show that K is locally compact), then (K, v) is a local field

(K, v) is a discretely valued field, we have $\widehat{\mathcal{O}_K}$ as a DVR with π again as the uniformizer, let $\hat{K}$ to be the field of fractions with a natural valuation $\hat{v}$, and K has dense image in $\hat{K}$. So if (K, v) has finite residue field, the completion is a local field

Example 5.5 $\mathbb{F}_q(t)$, v_t valuation, with residue field $\mathbb{F}_q$, valuation ring $\mathbb{F}_q[t]$, and max ideal $t\mathbb{F}_q[t]$, the completion is the Laurent series in t, v_t gives the order of zero or pole, with valuation ring $\mathbb{F}[[t]]$

K is a local field

Structure of $(K, +)$ and $(K^\times, \times)$

$K^\times \cong \mathbb{Z} \times \mathcal{O}_K^\times$ as topological groups, $\mathbb{Z}$ with discrete topology

$U = \mathcal{O}_K^\times$, $U_n = 1 + m_K^n$, then $U \supseteq U_1 \supseteq \cdots$ form an open subgroup neighborhood basis of 1, $U/U_1 \cong k^\times$, $U_n/U_{n+1} \cong m_K^n/m_K^{n+1} \cong k$, $x \mapsto x - 1$

U_1 is the unique pro-p Sylow subgroup of U since $|U_n/U_{n+1}| = p$ and $|U/U_1| = p - 1$ is coprime to p

U is profinite since U is compact(closed in $\mathcal{O}_K$)

Remark 5.6 If K is a local field of characteristic 0, then for sufficiently large n(so that the series converge), $m_K^n \cong U_n$, $x \mapsto e^x$

Corollary 5.7 In this case, every finite index subgroup of $K^\times$ is open

Proof. Suppose H is of finite index j in $K^\times$, then $(K^\times)^j \subseteq H$. Fix uniformizer π, $(K^\times)^j \cong \pi^{j\mathbb{Z}} \times U^j$. $U^j \supseteq U_1^j \supseteq \cdots$, $\underset{open}{}$ only need to show U_n^j is open for some n, for n large enough, $U_n \cong m_K^n \cong \mathcal{O}_K$, so $U_n^j \cong j\mathcal{O}_K \subseteq \mathcal{O}_K$

□

Teichmuller lift:

Fact 5.8 The surjective homomrophism $\mathcal{O}_K^\times \to k^\times$($k$ is the residue field which is finite) has a unique multiplicative section $[] : k^\times \to \mathcal{O}_K^\times$. Moreover, $[x] = \lim_{\underset{n}{\to}} y_n^{p^n}$, $y_n \in \mathcal{O}_K^\times$ is an arbitrary lift of $\sqrt{p^n}x \in k^\times$

Example 5.9 $K = \mathbb{Q}_5$, $[\bar{4}] = -1 \in \mathcal{O}_K^\times = \mathbb{Z}_5^\times$

Fact 5.10 $\forall x \in K^\times$, $\exists_1 (a_n)_{n \geq v(x)}$ such that $a_n \in \{0\} \cup [k^\times]$, $x = \sum_{n \geq v(x)} \pi^n a_n$

Warning: $x = sum\pi^n a_n$, $y = sum\pi^n b_n$, $x + y = x = sum\pi^n (a_n + b_n)$ is not the canonical choice

Finite extensions of local fields

Theorem 5.11 (Serre II.2) (K, v) is a complete discretely valued field, E/K is a separable field extension, $\exists_1 w$ on E and $\exists_1 e \in \mathbb{Z}_{\geq 1}$ such that $\forall x \in K$(e is the ramification index), $w(x) = ev(x)$. Moreover, (E, w) is complete, k_E/k_K is a finite extension of degree $f = [E : K]/e$

Remark 5.12 $w(y) = v(N_{E/K}(y))/f$, $\forall y \in E$

$\Rightarrow$ Every finite extension of a local field has canonical structure of a local field itself. In the future, when we talk about finite extensions of local fields E/K, it's always assumed that the local field structure on E is obtained from K in this way

Fact 5.13 Every local field is either a finite extension of $\mathbb{Q}_p$ or $\mathbb{F}_q([t])$, Laurent series

6 Galois theory for local fields

Definition 6.1 A finite separable extension E/K is unramified if $e(E/K) = 1$

Fact 6.2 If E/K is unramified, then it's Galois. $\mathrm{Gal}(E/K) \to \mathrm{Gal}(k_E/k_K)$ is an isomorphism

K is a local field

Fact 6.3 E/K finite unramified, L/K finite extension, $\mathrm{Hom}_K(E, L) \to \mathrm{Hom}_{k_K}(k_E, k_L)$ is a bijection. It's an equivalence of categories: unramified extensions over K to finite extensions over k_K

Thus we have

$$\{K \subseteq E \subseteq L, E/K \text{ unramified}\} \leftrightarrow \{\text{subextension of } k_L/k_K\}$$

K^s/K is the separable closure. $\forall n \in \mathbb{Z}_{\geq 1}$, $\exists_1 K \subseteq K_n \subseteq K^s$ such that K_n/K is unramified and of degree n. $K^{un} = \bigcup_{n\geq 1} K_n$, K^{un}/K is a Galois field extension and contains all possible finite unramified extensions of K inside K^s(all are of the form K_n). v on K extends to K^{un}, (K^{un}, v_{un}) is a discretely valued field(not complete in general), has $\widehat{K}$ for its completion

$\mathrm{Gal}(K^{un}/K) = \lim_{\leftarrow} \mathrm{Gal}(K_n/K) = \lim_{\leftarrow} \mathbb{Z}/n = \hat{\mathbb{Z}}$

Define the inertia group $I_K = \ker(\mathrm{Gal}(K^s/K) \to \mathrm{Gal}(K^{un}/K)) \cong \mathrm{Gal}(K^s/K^{un})$. For all separable finite extension E/K, let E_{ur} be the maximal unr ext of K inside E. $\mathrm{Gal}(E/E_{un}) = I(E/K)$, $\lim_{\leftarrow} \mathrm{Gal}(E/E_{ur}) = \lim_{\leftarrow} I(E/K) = I_K$

We have exact sequence

$$1 \to I_K \to G_K = \mathrm{Gal}(K^s/K) \to \mathrm{Gal}(K^{ur}/K) \cong \hat{\mathbb{Z}} \to 1$$

Exercise 6.4 $\mathrm{Gal}(K^{ab}/K)$ is the maximal Hausdorff and abelian quotient of G_K, i.e. $\ker(G_K \to \mathrm{Gal}(K^{ab}/K))$ is the closure of the derived subgroup of G_K

Theorem 6.5 (Local Reciprocity) $\exists_1 \psi_K : K^\times \to \mathrm{Gal}(K^{ab}/K)$ such that

1. For any uniformizer $\pi \in K$, $K^\times \to \mathrm{Gal}(K^{ab}/K) \to \mathrm{Gal}(K^{ur}/K)$ sends π to the Frobenius σ. Thus for any $x \in \mathcal{O}_K^\times$, x goes to the trivial element in $\mathrm{Gal}(K^{ur}/K)$, in fact, $\mathcal{O}_K^\times$ is the kernel
2. For all finite abelian extension L/K inside K^{ab}, $\psi_{L/K} : K^\times \to \mathrm{Gal}(K^{ab}/K) \to \mathrm{Gal}(L/K)$ has kernel $N_{L/K}(L^\times) \subseteq K^\times$ and is surjective and $\psi_{L/K} : K^\times/N_{L/K}(L^\times) \xrightarrow{\cong} \mathrm{Gal}(L/K)$

$$\begin{array}{ccccc} K^\times & \longrightarrow & \mathrm{Gal}(K^{ab}/K) & \longrightarrow & \mathrm{Gal}(K^{ur}/K) \\ & \searrow & & & \downarrow \\ & & \mathbb{Z} & \longrightarrow & \hat{\mathbb{Z}} \end{array}$$

Definition 6.6 By a norm subgroup of $K^\times$, we mean a subgroup of the fomr $N_{L/K}(L^\times)$ for some finite abelian extension L/K, write N_L for this to simplify notation

Remark 6.7 (Norm Limitation theorem) If L/K is an arbitrary finite separable extension, E/K is the maximal abelian extension of K inside L, then $N_L = N_E$

Corollary 6.8 L, L' are finite abelian extensions of K

1. $N_L \subseteq N_{L'} \Leftrightarrow L' \subseteq L$
2. $N_{LL'} = N_L \cap N_{L'}$

$$\begin{array}{ccccc} K^\times & \longrightarrow & K^\times/N_L & \longrightarrow & \mathrm{Gal}(L/K) \\ & \searrow & & & \downarrow \\ & & \mathrm{Gal}(K^{ab}/K) & \longrightarrow & \mathrm{Gal}(L'/K) \end{array}$$

Proof.

1. $N_{L/K} = N_{L/L'} N_{L'/K}$
2. $\mathrm{Gal}(LL'/K) \hookrightarrow \mathrm{Gal}(L/K) \times \mathrm{Gal}(L'/K)$

□

Corollary 6.9 Every norm subgroup is of finite index in $K^\times$. Every subgroup $K^\times$ containing some norm subgroup is itself a norm subgroup

Proof. Say $K^\times \supseteq M \supseteq N_L$, $K^\times/N_L \xrightarrow{\cong} \mathrm{Gal}(L/K)$, M/N_L corresponds to $\mathrm{Gal}(L/L')$ for some $L' \supseteq L$. So M is the kernel of $K^\times \to K^\times/N_L \to \mathrm{Gal}(L/K) \to \mathrm{Gal}(L'/K)$

□

Theorem 6.10 $L \to N_L$ is an inclusion reversing bijection So finite abelian extensions over K to normal subgroups

Lemma 6.11 L/K finite extension such that $N_{L/K}(L^\times)$ is of finite index in $K^\times$, then $N_{L/K}(L^\times)$ is open in $K^\times$. If $\mathrm{char}\ k = 0$, know every finite index subgroup of $K^\times$ is open

Proof.

□

7 3-2

$K_\pi = \cup_L L$, L finite alge ext of K such that $N_L = N_{L/K}(K^\times)$ containing π

Theorem 7.1 (local existence theorem) For every open finite index subgroup $U \subseteq K^\times$, there exists a finite abelian extsion L/K such that $N_L = U$

$K^{ab} = K_\pi K^{ur}$ is linearly disjoint

Remark 7.2 To proof this, we used the local existence theorem

In the proof, we have actually proved: Suppose H is a profinite group, $G = \mathbb{Z} \times H$ with the product topology, then every open finite index subgroup N of G contains $n\mathbb{Z} \times H'$, $H' \subseteq H$ is an open subgroup

Corollary 7.3 If $G = \mathbb{Z} \times H$, then $\hat{G} = \lim\limits_{\underset{n,H'}{\leftarrow}} G/n\mathbb{Z} \times H' = \lim\limits_{\underset{n,H'}{\leftarrow}} \mathbb{Z}/n \times H/H' = \hat{\mathbb{Z}} \times H$, since H is profinite

Theorem 7.4 Assuming the local existence theorem, then ϕ_K in the local reciprocity theorem must be unique

Proof. By local existence theorem, $K^{ab} = K_\pi K^{ur}$. Suppose ϕ, ϕ' are two candidates for ϕ_K, $\phi(\pi), \phi'(\pi)$ acts trivially on K_π, and acts as Frobenius on K^{ur}, then $\phi(\pi) = \phi'(\pi) \in \mathrm{Gal}(K^{ar}/K)$. But $K^\times$ is generated by all uniformizers

□

Idea of Lubin-Tate theory: Explicitly construct $K_{\pi,n}$

$K_{\pi,n} = \mathbb{Q}_p(\zeta_{p^n})$

$\mathbb{Q}_p(\zeta_{p^n}) = \mathbb{Q}_p(\Lambda_n)$

Λ_n is free of rank 1 over $\mathbb{Z}/p^n$

$\mathrm{Gal}(Q_p(\Lambda_n)/Q_p)$ permutes Λ_n, giving iso $\mathrm{Gal}(Q_p(\Lambda_n)/Q_p) \to (\mathbb{Z}/p^n)^\times$

8 3-4

Recall: want to construct $K_{\pi,n}$ as $K(\Lambda_n)$, $\Lambda_n = \{x \in \Lambda | [\pi^n]x = 0\}$, $\Lambda = \bigcup_L m_L$, L/K finite extension

$x +' y = x + y +$ higher power terms with coefficients in $\mathcal{O}_K$, $(\Lambda, +')$ also has $\mathcal{O}_K$-module structure, $a \in \mathcal{O}_K, x \in \Lambda$, $[a]x$

Formal group laws

A is a fixed commutative ring. A formal group low over A is a just $F \in A[[X,Y]]$, $F(X,Y) = X + Y +$ higher degree terms, satisfying axioms that mimic usual group axioms. (The difference is that we don't have a set)

Composition of power series

$f \in A[[x_1, \cdots, x_n]]$, $g_1, \cdots, g_n \in A[[y_1, \cdots, y_k]]$ want $f(g1(y1, ...yk), ..., gn(y1, ...yk))$. This doesn't work in general!

Counterexample: $n = k = 1$, $f(x) = 1 + x + x^2 + \cdots, g(x) = 1$, then $f(g(x))$ doesn't make sense. Need each g_i have no constant terms

$A[[y_1, \cdots, y_k]]_+$ is the set of g's without constant terms

$\forall f \in A[[x_1, \cdots, x_n]], g_1, \cdots, g_n \in A[[y_1, \cdots, y_k]]_+$, then $f(g_1, \cdots, g_n) \in A[[y_1, \cdots, y_k]]$, if $f \in A[[x_1, \cdots, x_n]]_+$, then $f(g_1, \cdots, g_n) \in A[[y_1, \cdots, y_k]]_+$

Example 8.1 $f(x) = g(x) = \sum_{n \geq 1} x^n$

Lemma 8.2 Let $f, g, h \in A[[x]]_+$,

1. this is associative
2. f has a right composition inverse $g \in A[[x]]_+$ iff the linear term of f is ax with $a \in A^\times$. When this is the case, it is also the left inverse

Remark 8.3 2. is a power series analogue of the "inverse function theorem"

Proof.

1. Obvious
2. $f = \sum a_i x^i$, $g = \sum b_i x^i$, if $x = f \circ g(x) = \sum a_i g^i$ with linear term $a_1 b_1 x$, hence $a_1 \in A^\times$. Conversely, if $a_1 \in A^\times$

□

Exercise 8.4 Formulate and prove a multi-variable version of 2.

Definition 8.5 A formal group low over A is $F \in A[[X,Y]]_+$(write $F(x,y) = x +_F y$) satisfying

1. Associative, i.e. $F(F(x,y),z) = F(x,F(y,z))$
2. Commutative(want it to be an abelian group), i.e. $F(x,y) = F(y,x)$
3. 0 is the identity, i.e. $F(x,0) = x$
4. Inverse, i.e. $\exists i(x) = -x +$ higher terms $\in A[[X]]_+$ such that $F(X, i(x)) = 0$

Remark 8.6 Have higher dimensional and non-commutative versions, we only care about the one dimensional commutative formal group laws

Observation: From 3. $F(x,y) = x + y + xyg$

Lemma 8.7 Suppose $F \in A[[x,y]]_+$ satisfies 1. and suppose $F(x,y) = x + y +$ higher terms, then it automatically satisfying the 3.

Proof. Set $f(x) = F(x,0) \in A[[x]]_+$ is of the form $x +$ higher terms, note that $F(0,0) = 0$, by associativity, $f \circ f(x) = F(F(x,0),0) = F(x,0) = f(x)$. Let g be the composition inverse of f, then $g \circ f \circ f(x) = g \circ f(x) = x$

□

Example 8.8 (Formal addtive group $\hat{\mathbb{G}}_a$) $F(x,y) = x + y$, $i(x) = -x$ works universal over arbitrary ring A

Example 8.9 (Formal multiplicative group $\hat{\mathbb{G}}_m$) $F(x,y) = x + y + xy = (x+1)(y+1) - 1$, $i(x) = \sum_{n\geq 1}(-x)^n$

Definition 8.10 Let $F, G \in A[[x,y]]_+$ be formal gropus laws, a homomorphism $f : F \to G$ is by definition an element of $A[[x]]_+$ such that $f(x +_F y) = f(x) +_G f(y)$, i.e. $f(F(x,y)) = G(f(x), f(y))$

Definition 8.11 $\mathrm{End}(F)$ is a ring under $+_F$ and composition

Example 8.12 $A = \mathbb{Q}$, $F(x,y) = x + y + xy$, $G(x,y) = x + y$, $f(x) = \log(1+x)$ is a homomorphims $F \to G$, $f(x +_F y) = f((1+x)(1+y) - 1) = \log(1+x) + \log(1+y) = f(x) +_G f(y)$

Let K be a local field, $\pi \in K$ a uniformizer, construct certain formal group laws over $A = \mathcal{O}_K$. These group laws are equipped with the structure of a formal A-module, i.e. $A \to \mathrm{End}(F)$

Definition 8.13 $\mathcal{F}_\pi$ is the set of $f \in A[[x]]_+$ such that

1. $f(x) = \pi x + O(x^2)$

2. $f(x) = \equiv x^q \mod \pi$, $\mathbb{F}_1 = \mathcal{O}_K/\pi$ i.e. $f(x) = \sum a_n x^n, a_n \in m_K, a_q \in 1 + m_K$

Example 8.14 $f(x) = \pi x + x^q$ $K = O_p, \pi = p, f(x) = (1+x)^p - 1$

Lemma 8.15 (Key lemma) $\forall f, g \in \mathcal{F}_\pi$, $\forall n \geq 1$, $\exists \phi \in A[[x_1, \cdots, x_n]]_+$ such that $f(\phi(x_1, \cdots, x_n)) = \phi(g(x_1), \cdots, g(x_n))$. Moreover, ϕ is uniquely determined by its linear part $a_1 x_1 + \cdots + a_n x_n$, also the linear part can be arbitrary

Proof. Start with $\phi_1 = a_1 x_1 + \cdots + a_n x_n$, construct uniquely by induction ϕ_r which is a degree r polynomial in $x_1, \cdots, x_n$ such that $f(\phi_r(x_1, \cdots, x_n)) = \phi_r(g(x_1), \cdots, g(x_n)) + O(x^{r+1})$ $\phi_r = \phi_{r-1} + Q_r$, where Q_r is homogeneous of deg r in $x_1, \cdots, x_n$ $f(\phi_1) = \pi\phi_1 + O(x^2)$, and $\phi_1(g) = \phi_1(\pi x_1, \cdots, \pi x_n) + O(x^2)$ Suppose ϕ_r has been constructed, $\phi_{r+1} = \phi_r + Q$ for some $\deg Q = r + 1$ $f(\phi_{r+1}) = f(\phi_r + Q) = f(\phi_r) + f'(\phi_r)Q + O(x^{2r+2}) = f(\phi_r) + \pi Q + O(x^{r+2})$ $\phi_{r+1}(g(x_1), \cdots, g(x_n)) = \phi_r(g(x_1), \cdots, g(x_n)) + Q(g(x_1), \cdots, g(x_n)) = \phi_r(g(x_1), \cdots, g(x_n)) + Q(\pi x_1, \cdots, \pi x_n) + O(x^{r+2})$ Set Q to be $\frac{[f(\phi_r) - \phi_r(g)] mod x^{r+1}}{\pi^{r+1} - \pi}$. Mod π, $f(\phi_r) \equiv \phi_r^q$, $\phi_r(g) \equiv \phi_r(x_1^q, \cdots, x_n^q) \equiv \phi_r^q$), so the numerator is divisible by π. Hence Q has coefficients in $A = \mathcal{O}_K$

□

Proposition 8.16 (Lubin-Tate formal group law) $\forall f \in \mathcal{F}_\pi$, $\exists_1$ formal group law $F_f = F(x,y) \in A[[x,y]]_+$ such that $f \in \mathrm{End}(F)$ which defines a Lubin-Tate group

Proof. By the Key lemma, there is a unique $F \in A[[x,y]]_+$ with linear part is $x + y$ such that $f(F(x,y)) = F(f(x), f(y))$, only need to check that F is a formal group law. Denote $G_1(x,y,z) = F(x, F(y,z)), G_2(x,y,z) = F(F(x,y), z)$, then $f(G_1(x,y,z)) = f(F(x, F(y,z))) = F(f(x), f(F(y,z))) = F(f(x), F(f(y), f(z))) = G_1(f(x), f(y), f(z))$, same for G_2, and they have the same linear parts $x + y + z$, so they are indeed equal

□

Remark 8.17 over $\mathbb{F}_q$, there is an endomorphism x^q

Fix $f \in \mathcal{F}_\pi$, ther is $\mathcal{O}_K \to \mathrm{End}(\mathcal{F}_f)$

Lemma 8.18 $\forall a \in \mathcal{O}_K, \forall f, g \in \mathcal{F}_\pi$, $exists_1$ homomorphism $[a]_{g,f} : F_f \to F_g$ which intertwines the endomorphisms f, g, i.e. $[a]_{g,f} \in \mathcal{O}_K[[x]]_+$, $[a]_{g,f}(F_f(x,y)) = F_g([a]_{g,f}(x), [a]_{g,f}(y))$, $[a]_{g,f}(f(x)) = g([a]_{g,f}(x))$, the linear part of $[a]_{g,f}$ is ax

Proof. By the key lemma, there exists a unique $h(x) \in \mathcal{O}_K[[x]]_+$ that intertwines f, g with linear part ax, only need to check h is a homomorphism, i.e. $F_g(h(x), h(y)) = h(F_f(x,y))$, note that bothe sides have linear part $ax + ay$

□

Lemma 8.19 $\forall a, b \in \mathcal{O}_K, \forall f, g, h \in \mathcal{F}_\pi$

1. $[a]_{g,f} +_{F_g} [b]_{g,f}$ is the of course the unique homomorphism $F_f \to F_g$ with linear part $(a+b)x$, i.e. $[a+b]_{g,f}$
2. $[b]_{h,g}[a]_{g,f} = [ab]_{h,f}$

Special case: $[1]_{g,f}, [1]_{f,g}, [1]_{f,f}, [1]_{g,g}$ are canonical isomorphisms $[]_{f,f} : \mathcal{O}_K \to \mathrm{End}(F_f)$ is a ring homomorhpism, and uniquely characterized by $[a]_{f,f}$ has linear part ax, $[\pi]_{f,f} = f$ We have a formal $\mathcal{O}_K$-module structure on F_f, $\mathcal{O}_K \to \mathrm{End}(F_f)$, $[1]_{f,g}$ is a formal $\mathcal{O}_K$ module isomorphism $F_f \to F_g$

Example 8.20 $K = \mathbb{Q}_p$, $\pi = p$, $f = (1+x)^p - 1 \in \mathcal{F}_p$, and $F_f(x,y) = (1+x)(1+y) - 1$. In this case, $[a]_{f,f} = (1+x)^a - 1$ for $a \in \mathbb{Z}_p$, why is $\binom{a}{i} \in \mathbb{Z}_p$, since $\mathbb{Z}$ is dense in $\mathbb{Z}_p$, let $a_k \to a$, $a_k \in \mathbb{Z}$, $\mathbb{Z} \ni \binom{a_k}{i} \to \binom{a}{i}$ in $\mathbb{Q}_p$, but $\mathbb{Z}_p$ is closed in $\mathbb{Q}_p$, to finish, see $(1+x)^a(1+y)^a - 1 = ((1+x)(1+y))^a - 1$, and $[a]_{f,f}$ commutes with f

Extending valuation $v : K \to \mathbb{Z} \cup \{\infty\}$ to $\bar{K}$

recall L/K finite ext, $\exists_1$ discrete valuation $w : L \to \mathbb{Z} \cup \{\infty\}$, recall w must be surjective, $w(x) = e(L/K)v(x)$, so we can extend and define $v : L \to \mathbb{Q} \cup \{\infty\}$, $v(y) = w(y)/e(L/K)$, in the end have $v : \bar{K} \to \mathbb{Q} \cup \{\infty\}$

Define $\Lambda = \left\{x \in \bar{K} | v(x) > 0\right\}$, $\Lambda = \cup_L m_L$, L runs over finite ext

Obeservation: $\forall f(x_1, \cdots, x_n) \in \mathcal{O}_K[[x_1, \cdots, x_n]]$ with constant term in Λ(resp m_L), $\forall x_1, \cdots, x_n \in \Lambda$(resp m_L), $f(x_1, \cdots, x_n)$ converges in Λ(resp m_L). For example $f(x) = a_0 + a_1 x + \cdots, x \in \Lambda$, so $x \in m_L$ for some L, just need $a_i x^i = 0$, but $v(a_i x^i) = v(a_i) + iv(x)$

For $f \in \mathcal{F}_\pi$, we can make Λ into a group Λ_f with addition $+_{F_f}$, and an $\mathcal{O}_K$ module $[a]_{f,f}$

9 3-23

Definition 9.1 For any A module M, $M_n = \ker(M \xrightarrow{\pi^n} M)$, this is independent of the choice of π, consider $\Lambda_{f,n} \subseteq \bar{K}$, define $K_{\pi,n}$ to be $K(\Lambda_{f,n})$ which is independent of the choice of f (reason: $x \in \Lambda_{f,n} \cap L$ for some L/K finite, $[1]_{g,f} : \Lambda_{f,n} \to \Lambda_{g,n}$ is an isomorphism since it is an isomorphism between F_f and F_g, so $[1]_{g,f}(x) \in \Lambda_{g,n} \cap L$, since L can be arbitrary, and the smallest field extension containing x is unique)

Claim: $K_{\pi,n} = K(rootsof f^{(n)})$, $f(x) = X^q + \pi x$, $f^{(n)}(x) = f^{\circ n}$

Newton Polygon[Neukirch's book]: Suppose $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in K[x]$, (K, v) disc valued field draw $(i, v(a_i))$, join them so that it the joining curve is convex(the convex hull) i-th segment has slope s_i, horizontal dist n_i, $\sum n_i = n$. Among all the roots of f in $\bar{K}$(count with multiplicity), exactly n_i of them have valuation -s_i

$f^{(n)}(x) \equiv x^{q^n} \mod \pi$ and $f^{(n)}(x) = x^{q^n} + \cdots + \pi^n x$. Try the Newton Polygon, exactly one with valuation ∞ which corresponds to 0, the others have positive valution ≥ 1, justifying the claim So $K_{\pi,n}/K$ is Galois, being the splitting field of $f^{(n)}(x)$

Lemma 9.2 For any $f \in F_\pi$, the module $\Lambda_{f,1}$ has q distinct elements. The map $\pi : \Lambda_f \to \Lambda_f$ is surjective

Proof. f and f' are coprime, so f has q distinct roots. And use Newton polygon, for any $\lambda \in \Lambda$ see $f(x) - \lambda = x^q - \pi x - \lambda$ has roots of valuations ≥ 1

□

Lemma 9.3 $K_{\pi,n}/K$ is separable

Proof. Induction on n. $n = 1$, $K_{\pi,1} = K(\Lambda_{f,1})$ is true by Lemma 9.2. For $\alpha \in \Lambda_{f,n+1}$, need to show α is separable over K. $f(\alpha) = \pi \cdot_\phi \alpha \in \Lambda_{f,n}$, denote as β, α is a root of $g(x) = f(x) - \beta$, $g' = f' = qx^{q-1} + \pi$, all roots of g' has valuation ≤ 0, hence α is not a root of g', so α is a simple root of g, and $g \in K_{\pi,n}[x]$

□

Proposition 9.4 $\forall f \in F_\pi$ and $n \geq 1$, the A-module $\Lambda_{f,n}$ is iso to $A/\pi^n A$. In particular, it has q^n elements

Proof. First show $|\Lambda_{f,n}| = q^n$. We have short exact sequence

$$0 \to \Lambda_{f,1} \to \Lambda_{f,n+1} \xrightarrow{\pi} \to \Lambda_{f,n} \to 0$$

The surjectivity is due to Lemma 9.2. By classification theorem of finitely generated module over a PID(A)

$$\Lambda_{f,n} \cong A/\pi^{n_1} \oplus \cdots \oplus A/\pi^{n_k}$$

So the π torsion of $\Lambda_{f,n}$ is $\pi^{n_1-1}A/\pi^{n_1} \oplus \cdots \oplus \pi^{n_k-1}A/\pi^{n_k} \cong A/pi \cdots \oplus A/\pi$, k copies, which is impossible since it should be $\Lambda_{f,1}$

□

As a consequence, the group of A-module automorphisms of $\Lambda_{f,n}$ is equal to $(A/\pi^n)\hat{} \times$

$\mathrm{Gal}(K_{\pi,n}/K)$ acts on $K_{\pi,n}$, and $x \in \Lambda_{f,n}$ iff x is a root of $f^{(n)}$ which is acted on by the Galois group, f can be power series in $A[[x]]_+$ since the action is continuous and preserve the valuation. In fact, $\mathrm{Gal}(K_{\pi,n}/K)$ acts on $\Lambda_{f,n}$ preserving the A-module structure, so we get the lubin-Tate homomorphism $\rho_{\pi,n} : \mathrm{Gal}(K_{\pi,n}/K) \to (A/\pi^n)^\times$

Theorem 9.5

1. $K_{\pi,n}/K$ is totally ramified with deg $(q-1)q^{n-1}$

2. $\rho_{\pi,n}$ is an isomorphism

3. π is a norm from $K_{\pi,n}$, i.e. $\exists y \in K_{\pi,n}$, $N_{K_{\pi,n}/K}(y) = \pi$

Proof.

1. Take a non-zero root $\pi_1 \in \Lambda_{f,1} - \{0\}$ of f. Take $\pi_2 \in \Lambda_{f,2} - \Lambda_{f,1}$ to be a root of $f(x) - \pi_1$, inductively defined $\pi_i \in \Lambda_{f,i} - \Lambda_{f,i-1}$'s. $v(\pi_1) = \frac{1}{q-1}$, use Newton polygon, $v(\pi_2) = \frac{1}{(q-1)q}$, and $v(\pi_n) = \frac{1}{(q-1)q^{n-1}}$, hence the ramification index $e(K_{\pi,n}/K) \geq (q-1)q^{n-1}$. Since $\rho_{\pi,n}$ is obviously injective, $[K_{\pi,n} : K] \leq |(A/\pi^n A)^\times| = (q-1)q^{n-1}$

2. Also done by 1

3. $f^{(n-1)}(\pi_n) = \pi_1$, and $\frac{f(x)}{x} \circ f^{(n-1)}(\pi_n) = 0$, denote the polynomial h, then $\deg h = (q-1)q^{n-1}$, hence the monic minimal polynomial, justifing 3, $K_{\pi,n} = K(\pi_n)$, $N_{K_{\pi,n}/K}(\pi_n) = (-1)^{(q-1)q^{n-1}}\pi$, and the sign is 1 unless $n = 1$ and $q = 2^k$

□

It turns out, the maximal abelian extension $K^{ab} = K_\pi \cdot K^{ur}$(linearly disjoint, since K_π is totally ramified), $K_\pi = \cup_{n\geq 1} K_{\pi,n}$. Hence $\mathrm{Gal}(K^{ab}/K) = \mathrm{Gal}(K_\pi/K) \times \mathrm{Gal}(K^{ur}/K) \cong \mathrm{Gal}(K_\pi/K) \times \hat{\mathbb{Z}}$

Recall the Lubin-Tate homomorphism $\rho_{\pi,n} : \mathrm{Gal}(K_{\pi,n}/K) \to (A/\pi^n)^\times$ is an iso, taking inverse limit, we have iso $\rho_\pi : \mathrm{Gal}(K_\pi/K) \to A^\times$. So $\mathrm{Gal}(K^{ab}/K) \cong A^\times\hat{\mathbb{Z}}$. It will turn out that the local reciprocity map $\phi : K^\times \to \mathrm{Gal}(K^{ab}/K)$ is given by $K^\times = \pi^Z \times A^\times \to \hat{Z} \times A^\times$, $(\pi^i, a) \mapsto (i, a^{-1})$

Definition 9.6 $K^{LT} = K_\pi K^{ur}$, $\phi_\pi : \mathrm{Gal}(K^{LT}/K)$(don't yet know that $K^{LT} = K^{ab}$)

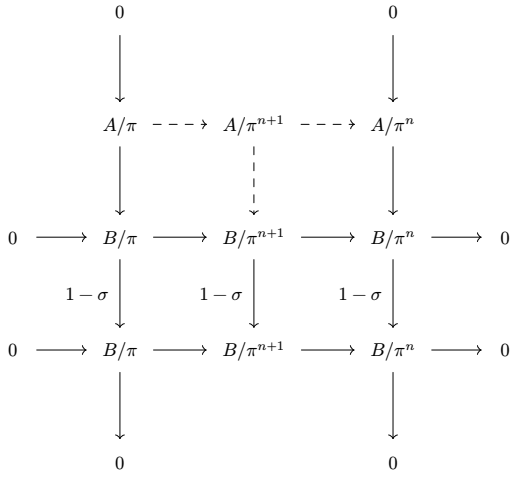
Lemma 9.7

1. $(B, +) \to (B, +)$, $x \mapsto x - \sigma(x)$, is surjective, and kernel is A

2. $B^\times \to B^\times$, $x \mapsto x \cdot \sigma(x)^{-1}$ is surjective and kernel is $A^\times$

Proof.

1. Prove by induction that $0 \to A \to B \xrightarrow{1-\sigma} B \to 0$ is exact, so we prove $0 \to A/\pi^n \to B/\pi^n \to B/\pi^n \to$ is exact. The case $n = 1$, it's just $0 \to F_q \to \bar{F}_q \to \bar{F}_q \to 0$ is exact by Galois theory. Assuming exactness for n, for $n+1$, $0 \to B/\pi \to B/\pi^{n+1} \to B/\pi^n \to 0$



   Then use Snake lemma. In general, the inverse limit is only left exact, but recall $0 \to A_i \to B_i \to C_i \to 0$ are short exact sequences, Mittag-Leffler condition: $\forall i$, the subgroups $\mathfrak{I}(A_j \to A_i)$ of A_i stablizes when j is large, the the inverse limit is exact. Here $B = \lim_{\leftarrow} B/\pi^n$ since B is complete

2. Multiplicative analogue

□

Proposition 9.8 $\pi, \varpi = \pi u$, $f \in F_\pi, g \in F_\pi$. Fix $\epsilon \in B^\times$ such that $\sigma(\epsilon)\epsilon^{-1} = u \in A^\times$, by surjection $B^\times \to B^\times$, i.e. $\sigma(\epsilon) = u\epsilon$. There exists $\theta(T) \in B[[T]]_+$ such that

1. the linear coefficient of θ is ϵ
2. $\sigma\theta = \theta \circ [u]_f$
3. θ is an iso $F_f \to F_g$(base changed from A to B)
4. θ is A-linear, i.e. $\forall a \in A$, $[a]_g \circ \theta = \theta \circ [a]_f$. Thus θ is an iso between formal A-modules F_f and F_g over B

Proof. Construct θ satisfying $1, 2$ by applying Lemma. Given such θ, let's modify it so it also satisfies $3, 4$. Actually, we modify θ such that $g = \sigma\theta \circ f \circ \theta^{-1} = \theta \circ [u]_f \circ f \circ \theta^{-1} = \theta \circ f \circ [u]_f \circ \theta^{-1} = h$, $\sigma h = \sigma\theta \circ f \circ [u]_f (\sigma\theta)^{-1} = \sigma\theta \circ f \circ \theta^{-1} = h$, so $h \in A[[x]]_+$

□

Theorem 9.9 K^{LT}, ϕ_π is independent of π

Proof. Idea: If π, ϖ are two different uniformizers, $f \in F_{\pi,,}$ $f \in F_{\varpi,,}$ they are iso whose base change from $A = O_K$ to $B = O_{\breve{K}}$, $\breve{K}$ is the completion of K^{ur}

□

Fact 9.10 If L is discretely valued field, then L is algebraically closed in $\hat{L}$, i.e. every element in $\hat{L} - L$ is transcendental over L

Fact 9.11 Suppose L/K is a finite extension of discretely valued fields, then K is algebraically closed in $\hat{L}$

Frobenius reciprocity:

$$\mathrm{Hom}_G(X, CoInd_H^G A) \cong \mathrm{Hom}_H(X, A)$$

$$\mathrm{Hom}_G(Ind_H^G A, X) \cong \mathrm{Hom}_H(A, X)$$

Lemma 9.12 $\mathrm{Ind}^G : Ab \to G - modules$ takes inj to inj, $CoInd^G$: takes proj to proj

Corollary 9.13 The category of G modules have enough inj and proj

A G module X is relatively inj if it's a direct summand of a coinduced G module. relative proj.

Proposition 9.14 $H^i(G, X) = 0, \forall i \geq 1$ For X relative inj. $H_i(G, Y) = 0, \forall i \geq 1$ For Y relative proj

Proof. $P_\bullet \to \mathbb{Z} \to 0$ free resolution. Without loss of generality, say $X = CoInd^G A$, $\mathrm{Hom}_G(P_\bullet, CoInd^G A) \cong \mathrm{Hom}_Z(P_\bullet, A) = 0$

□

10 Tate cohomology

G is finite, the goal is to put homology and cohomology in the same framework

Fact 10.1 If P is relatively projective module, then $\hat{H}^i(G, P) = 0$ for all i

Cup product for Tate cohomology $A \otimes_Z B$ is a G module with diagonal action of G

The cup product $\hat{H}^p(X, A) \otimes \hat{H}^q(X, B) \to \hat{H}^{p+q}(X, A \otimes B)$ is uniquely characterized by

1. It is functorial in A and B
2. For $p = q = 0$, it is induced by $A^G \otimes B^G \to (A \otimes B)^G$, $a \otimes b \mapsto a \otimes b$
3. compatible with connecting homomorphism If

$$0 \to A \to A' \to A'' \to 0$$

is exact such that

$$0 \to A \otimes B \to A' \otimes B \to A'' \otimes B \to 0$$

is exact, the $\forall \beta \in \hat{H}^q(B)$

$$\begin{array}{ccc} \hat{H}^p(A'') & \xrightarrow{- \smile \beta} & \hat{H}^p(A'' \otimes B) \\ \downarrow{\scriptstyle \delta} & & \downarrow{\scriptstyle \delta} \\ \hat{H}^{p+1}(A) & \xrightarrow{- \smile \beta} & \hat{H}^{p+1}(A \otimes B) \end{array}$$

Similar for $\alpha \smile -$, up to a sign $(-1)^p$ which comes from the fact that it is graded commutative

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